MILITARY SPECIFICATION

BORON, AMORPHOUS, POWDER

1. SCOPE

1.1 This specification covers amorphous boron powder for use in the manufacture of pyrotechnics.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-D-723 - Drums, Fiber

MILITARY

JAN-P-112 - Packaging and Packing for Overseas Shipment Drums, Plywood (for drums whose weight of contents does not exceed 200 pounds).

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Inspection Terms and Definitions.
MIL-STD-129 - Marking of Shipments.

PUBLICATIONS

ORDNANCE CORPS

ORD-M608-11 - Procedures and Tables for Continuous Sampling by Attributes.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

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3. REQUIREMENTS

3.1 Composition.- The composition of the amorphous boron powder shall be as follows, when determined as specified in 4.3.

	Percent
Amorphous boron	84.0, minimum (min.)
Insoluble in Hydrogen Peroxide	1.5, maximum (max.)
Total magnesium	8.2, max.

3.2 Moisture content.- The moisture content shall be 0.5 percent max.

3.3 Average particle diameter.- The average particle diameter shall be 1.0 microns, max., when determined as specified in 4.3.5.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.- The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein. Inspection shall be performed in accordance with this specification and other specifications referenced in any of the contractual documents.

4.1.1 Contractor quality assurance system.- If the contractor desires to utilize a quality assurance system, which is at variance with the quality assurance provisions of 4.2 and 4.3 and other documents referenced herein, he shall submit a written description of the system to the contracting officer for approval prior to initiation of production. It shall include a description covering controls for lot formation and identification, inspections to be performed, inspection stations, sampling procedures, methods of inspection (measuring and testing equipment), and provisions for control and disposition of non-conforming material. The written description will be considered acceptable when, as a minimum, it provides the quality assurance provisions required by the provisions of 4.2 and 4.3 and the other documents referenced herein. The contractor shall not be restricted to the inspection station or the method of inspection listed in this specification provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether certain procedures of the contractor's system provide equal assurance, the comparable procedure of this specification shall apply. The contractor shall notify the Government of, and obtain approval for, any changes to the written procedure that effects the degree of assurance required by this specification or other documents referenced herein.

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4.1.2 Submission of product.- At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

b. Quantity of product inspected.

c. Results obtained for all inspection performed.

d. Specification number and date, together with an identification and date of changes.

e. Certificates of analysis on all material, referred to by Government specifications, procured directly by the contractor.

f. Quantity of product in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 Government verification.-:Using the contractor's written quality assurance procedure (see 4.1.1), this detailed specification, and other contractual documents as a guide, the Government inspector shall verify all quality assurance operations performed by the contractor. Verification shall be in accordance with a or b as applicable, the decision being the responsibility of the procuring activity. In either case, the inspector shall also ascertain, prior to acceptance, that all quality provisions of other specifications referenced in any of the contractual documents have been complied with. Deviations from prescribed or agreed upon procedures discovered by the Government inspector shall be brought to the attention of the supplier. Disposition of the product and remedial action shall be as directed by the Government inspector, and depending on the nature of the deviation, may consist of lot rejection, screening, re-sampling, re-instruction of the supplier's employees, or other appropriate action:

a. Verification at the point of manufacture shall be accomplished at unscheduled intervals in accordance with 4.1.3.1 and 4.1.3.2.

b. Verification at the point of delivery shall be in accordance with 4.1.3.2.

4.1.3.1 Surveillance.- Surveillance shall include, but is not limited to:

a. Observation of procedures concerning lot formation and identification.

b. Observation of sampling procedures and application of acceptance criteria.

c. Determination that all required examinations and tests are performed in accordance with the prescribed procedures of this specification, or approved equivalents thereto.

d. Review of procedures for control and disposition of non-conforming material.

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4.1.3.2 Product inspection.-Product inspection shall consist of Government inspection of product which has been previously inspected by the contractor and found to meet the quality assurance provisions of this specification. The inspection by the Government shall be performed in order to determine that the product is of the quality required by this specification and that the contractor's records are reliable.

4.2 Inspection provisions.

4.2.1 Lot formation.- A lot shall consist of amorphous boron from the same blending operation from one manufacturer, in accordance with the same specification or same specification revision, under one continuous set of operating conditions. Each batch shall consist of that quantity of boron amorphous powder that has been subjected to the same unit chemical or physical process intended to make the final product homogeneous.

4.2.2 Examination.- Sampling plans and procedures for the following classification of defects shall be in accordance with Standard MIL-STD-105. Continuous sampling plans, in accordance with Handbook ORD-M608-11 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect.

4.2.2.1 Drum (prior to filling) (polyethylene liner)

Categories	Defects	Method of inspection
Critical: None defined.		
Major:	AQL 0.25 percent	
	101. Liner cut, torn or punctured	Visual
Minor: None defined.		

4.2.2.2 Drum (prior to closing filled container)

Categories	Defects	Method of inspection
Critical: None defined.		
Major:	AQL 0.25 percent	
	101. Bag liner missing or torn	Visual
	102. Bag liner improperly sealed	Visual
Minor:	AQL 0.40 percent	
	201. Bag liner pulled too tight, not allowing slack	Visual

4.2.2.3 Drum (after closing container)

Categories	Defects	Method of inspection
Critical: None defined.		

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Major:	AQL 0.40 percent	
101.	Weight content	Scale
102.	Sealing strips missing	Visual
103.	Marking missing, incorrect or illegible	Visual
104.	Closure incomplete or damaged to the extent that contents sift out	Visual
Minor:	AQL 1.00 percent	
201.	Sealing strips not properly applied	Visual
202.	Marking not properly applied	Visual

4.2.3 Testing.

4.2.3.1 Sampling by lot.- A random sample of 10 containers shall be selected from each lot. When lots are comprised of 10 containers or less each container shall be sampled.

4.2.3.1.2 Preparation of composite.- Approximately two ounce primary samples of Boron shall be removed from each of the ten containers in order to equal twenty ounces, and if there are less than ten containers, equal primary samples in sufficient quantity to equal twenty ounces, shall be removed from each container. The individual primary samples shall than be combined in order to form a homogeneous composite sample of twenty ounces and subjected to the tests specified in 4.3. If the composite sample fails to comply with any of the requirements specified, the lot shall be rejected.

4.3 Test methods and procedures.- The following test methods and procedures shall be performed.

4.3.1 Amorphous boron.

4.3.1.1 Total boron.- A sample of 0.10 to 0.15 grams (gm.) shall be accurately weighed in a tared platinum crucible. To the crucible 1.5 gm. of sodium carbonate shall be added and mixed well with a thin glass rod, taking care that no part of the mixture adheres to the rod. The crucible shall be covered and heated with a low flame of a Fisher burner for approximately five minutes, taking care to avoid loss by spattering or frothing. The heat shall be gradually increased to fuse the mixture and held at fusion temperature for five minutes. The mixture shall be cooled and then dissolved by placing the crucible and cover in a covered 400 milliter (ml.) beaker containing 50 ml. of 1-1 hydrochloric acid. The crucible and cover shall be removed and washed when digestion is complete , combining the washings with the solution. The solution shall be transferred to a 250 ml. volumetric flask and diluted to the mark. A 50 ml. aliquot shall be pipetted into a 250 ml. Erlenmeyer flask, three drops of methyl red indicator shall be added and the excess hydrochloric acid neutralized with 6 Normal (N.) sodium hydroxide. The pH shall be adjusted to the methyl red neutral point with 0.1 N. sodium hydroxide or 0.1 hydrochloric acid, whichever is necessary, and then 5 ml. of 0.1 N. hydrochloric acid added. A few glass beads shall be added, the solution boiled for three minutes to expel carbon dioxide, cooled in a water bath, and again neutralized to the methyl red neutral point with carbon dioxide free 0.1 N sodium hydroxide. Six to ten gm of reagent grade mannitol

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-and 20 drops of phenolphthalein indicator shall be added and titrated with 0.1 N sodium hydroxide to the first phenolphthalein pink coloration. A flask with the reagents above shall be run following the above procedure exactly. The percentage total Boron shall be calculated as follows:

$$\text{percent total Boron} = \frac{(A - B)(N)(1.082)}{W}$$

where:

A = ml. of sodium hydroxide used for sample.
B = ml. of sodium hydroxide used for blank.
N = normality of sodium hydroxide.
W = Weight of sample in aliquot.

4.3.1.2 Water soluble boron.- Approximately 0.500 gm. of sample shall be accurately weighed and transferred to a 400 ml. beaker. Approximately 200 ml. distilled water shall be added and the suspension placed under mechanical agitation. The pH shall be adjusted with 0.1 N sodium hydroxide or hydrochloric acid to 5.4 using a glass-calomel electrode pair and a potentiometer, approximately 10 gm. of dry mannitol shall be added and the solution titrated with 0.1 N sodium hydroxide to pH 8.0. A blank shall be run using the above procedure on equal amounts of reagent. The water soluble Boron shall be calculated as follows:

$$\text{percent water soluble Boron} = \frac{(A - B)(N)(1.082)}{W}$$

where:

A = ml. of sodium hydroxide used for sample.
B = ml. of sodium hydroxide used for blank.
N = normality of sodium hydroxide.
W = weight of sample.

4.3.1.3 Amorphous boron.- The amorphous shall be calculated as follows:

$$\text{percent Amorphous boron} = A - B$$

where:

A = percent total Boron
B = percent water soluble Boron.

4.3.2 Insoluble in hydrogen peroxide.- Approximately 0.3 gm. of the sample shall be accurately weighed and placed in 400 ml. beaker. The sample shall be moistened with a small quantity of water (1-2 ml.) and then 50 ml. of 30 percent hydrogen peroxide and 10 to 15 drops of concentrated nitric acid shall be added. After the initial relatively violent reaction has subsided, the solution shall be boiled for one hour, cooled, and filtered

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through a tared fine sintered glass crucible. Retain the filtrate and washings for the magnesium determination. The insoluble material shall be washed with hot water (100-200 ml.) dried at 100 degree centigrade (C.) cooled and weighed. The insoluble in hydrogen peroxide shall be calculated as follows:

$$\text{percent insoluble in hydrogen peroxide} = \frac{100\ A}{W}$$

where:

A = weight of insoluble material
W = Weight of sample.

4.3.3 Magnesium.- The filtrate and washing from 4.3.2 shall be transferred to a 400 ml. beaker. The solution shall be stirred at a moderate rate and 15 ml. of a 25 percent solution in diammonium phosphate and 1 ml. 2,4 - dinitrophenol indicator solution shall be added. The solution shall be neutralized by the dropwise addition of concentrated ammonium hydroxide, followed by the addition of a 10 ml. excess of ammonium hydroxide. The beaker shall be covered with a watch glass and the contents stirred for one hour. The beaker shall then be placed in a refrigerator (approximately 10 degrees C.)for a minimum of three hours, preferably overnight. The contents of the beaker shall be filtered through a Whatman number 42 or equivalent filter paper and the precipitate washed thoroughly with a wash solution containing 80 ml. 25 percent diammonium phosphate solution, 300 ml. distilled water and 20 ml. concentrated ammonium hydroxide. The filtrate and washing shall be discarded. The precipitate shall be dissolved by pouring 50 ml. of hot 1:20 hydrochloric acid through the filter, collecting the solution in a 250 ml. beaker. While stirring, all 0.5 ml. of 25 percent diammonium phosphate solution and 1 ml. of 2,4 - dinitrophenol indicator solution to the beaker. The solution shall be neutralized by the dropwise addition of concentrated ammonium hydroxide, followed by the addition of 4.0 ml. excess ammonium hydroxide. The beaker shall be covered with a watch glass and stirred for 10 minutes. The beaker shall be placed in a refrigerator for a minimum of three hours, preferably overnight. The contents of the beaker shall be filtered through a Whatman number 42 filter paper or equivalent and the precipitate washed with cold 1:20 ammonium hydroxide until the washings are phosphate free, as shown by a clear solution when the washings are added to a calcium chloride solution. The precipitate and paper shall then be placed in a preignited, tared porcelain crucible and dried in a 110 degree C. oven. The precipitate shall then be ignited, charring the paper over a low flame or a muffle furnace at first, then heating moderately until all the carbon is oxidized and finally igniting the precipitate at approximately 1000 degrees C. until constant weight is attained. The percent magnesium shall be calculated as follows:

$$\text{percent magnesium} = \frac{(A)\ 21.84}{W}$$

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where:

A = gain in weight of crucible
W = weight of sample.

4.3.4 Determination of moisture content.- An accurately weighed portion of approximately 5 gm. of the sample shall be transferred to a tared, glass stoppered weighing bottle about 60 millimeters in diameter. The weighing bottle and contents shall be placed, with the stopper removed, in an oven maintained at 100° to 105°C., for 2 hours. The stopper shall be replaced, the bottle cooled in a desiccator, and weighed. The moisture content shall be calculated as follows:

$$\text{Percent moisture} = \frac{(A)\ 100}{W}$$

where:

A = loss in weight of sample.
W = weight of sample.

4.3.5 Determination of average particle diameter.- The average particle diameter shall be determined using a Fisher subsieve sizer. One of the portion plugs shall be attached to the plug manipulator. A paper disk shall be laid over one end of the sample tube, and the plug pushed into the sample tube for a distance of approximately 0.5 inch, with the perforated surface of the plug against the surface of the paper disk, forcing the paper to crimp around the edges and precede the plug into the sample tube. The sample tube shall be placed in a vertical position with the paper side of the plug up. A 2.45 gm. portion of the dry sample shall be accurately weighed and transferred into the sample tube with the aid of a small funnel. The side of the tube shall be tapped to settle the powder. A second paper disk shall be laid over the top of the sample tube, the second porous plug attached to the plug manipulator, and the plug and paper disk forced down into the sample tube, manually compressing the sample as tightly as possible. The manipulator shall be removed. The rack shall be lowered until the pointer lies on the base line. The calculator chart shall be shifted until a porosity value of 0.80 is indicated by the pointer on the porosity scale located along the bottom edge of the calculator chart. The porosity value is a measure of the degree of packing of the sample. After making this setting, the chart shall not be moved until the determination has been completed. The rack shall be raised and the sample tube placed on the brass post beneath the rack and pinion with the lower plug in contact with the upper end of the brass post. The rack shall be lowered until the flat bottom end comes in contact with the upper plug. The rack shall continue to be lowered by turning the pinion knob manually until the tip of the pointer coincides with a point on the sample height curve on the chart. The sample has now been packed to a porosity value of 0.80. The rack shall be raised and the sample tube removed. The initial level of the water meniscus in the manometer tube, located over the calculator chart, shall be adjusted

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by means of the manometer control knob so that the meniscus coincides with the upper edge of the metal crossbar attached on the rack when the tip of the pointer coincides exactly with the base line on the calculator chart. If adjustment of the water meniscus cannot be made because of too little water in the manometer tube, water may be added through the top of the manometer tube. The range control knob, located at the extreme upper right of the front panel, shall be turned to the "LO" position. The sample tube shall be mounted, without disturbing the sample in any way, between the rubber-cushioned supports of the sample tube holder. The upper cap shall be screwed down onto the sample until an air tight seal is obtained at both ends. The line cord shall be plugged into a 110-volt, 60-cycle alternating current line, and the electrical switch at the lower right-hand corner of the front panel thrown to the "ON" position. This turns on the air pump as well as the pilot lamp which illuminates the tip of the bubbler tube in the pressure regulator stand-pipe, as observed through the round window in the lower left-hand corner of the front panel, and the level of water in the standpipe, as observed through the upper window. The pressure control knob shall be adjusted until the bubbles rise in the standpipe at the rate of 2 to 3 per second. The water level in the manometer tube shall be allowed to rise to a maximum level. Since the bulk volume of the required weight of sample is too great to permit the use of the calculator chart furnished with this apparatus, the average particle diameter shall be calculated using the following formula:

$$\mu = \frac{K\,H\,N}{(A\,H-N)3/2} \cdot \sqrt{\frac{F}{P-F}}$$

where:

μ = average particle diameter, microns
K = permeability constant
A = cross-sectional area of sample tube, square centimeters (cm.)
H = height of sample, cm.
F = height of water column in manometer tube, cm., when the sample tube contains plugs, filter disks, and sample.
P = height of water column in manometer tube, cm., when the sample tube contains plugs and filter disks only.
N = a factor which is equivalent to the weight of the sample divided by its true density.

The above determination shall be repeated at lower porosities, taking care to readjust the chart as required, until the bed is tightly compressed. All these determinations shall be made on the same sample. The porosities shall be plotted against the corresponding values of average particle diameter. Record the average particle size for which a change in porosity no longer affects the average particle diameter.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 Level A.- Unless otherwise specified in the contract or purchase order, Boron powder shall be packed in a fiber drum or plywood drums in

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accordance with Specification PPP-D-723 (Type III Grade A), JAN-P-112. The drums shall be furnished with a nominal 0.004 inch thick polyethylene liner, properly heat sealed or otherwise closed to afford protection to the contents equivalent to that given by a heat seal closure. The drums shall be closed in accordance with Specifications PPP-D-723 and JAN-P-112 as applicable.

5.1.2 Level C.- Boron powder shall be packed to insure acceptance by common or other carriers for safe transportation, at the lowest rate to the point of delivery.

5.2 Marking.- Containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following: a) title, number and date of this specification. b) Level of packing required.

6.2 Intended use.- The intended use for the Amorphous boron in this specification is a fuel in delays, non-gaseous fuzes, and tracers.

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:
Army-Ordnance Corps

Preparing activity:
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